United States Patent Application Publication 20250280002
Kind Code A1
Publication Date September 04, 2025
Inventor(s) YAMANAKA; Yuki et al.

COMMUNICATION CONTROL SYSTEM, GATEWAY, AND COMMUNICATION CONTROL METHOD

Abstract

A communication control system includes a gateway and an IoT device connected to the gateway, wherein the gateway includes a request module configured to request, from the IoT device, information regarding normal communication performed by the IoT device, and a creation module configured to create a whitelist in which details of communication permitted for each IoT device are designated on a basis of the information regarding normal communication performed by the IoT device, which is transmitted from the IoT device, and the IoT device includes a transmission module configured to transmit the information regarding normal communication performed by the IoT device to the gateway in response to a request from the request module.

Inventors: **YAMANAKA; Yuki (Tokyo, JP), TAKIGUCHI; Hiroyoshi (Tokyo, JP), SHINOHARA; Masanori (Tokyo, JP), MINAMI; Takuya (Tokyo, JP), WADA; Yasunori (Tokyo, JP), KANEMOTO; Yo (Tokyo, JP), NOMURA; Hiroto (Tokyo, JP)**

Applicant: **NIPPON TELEGRAPH AND TELEPHONE CORPORATION** (Tokyo, JP)

Family ID: **91084054**

Assignee: **NIPPON TELEGRAPH AND TELEPHONE CORPORATION (Tokyo, JP)**

Appl. No.: **19/203157**

Filed: **May 08, 2025**

Related U.S. Application Data

parent WO continuation PCT/JP2022/042443 20221115 PENDING child US 19203157

Publication Classification

Int. Cl.: **H04L9/40** (20220101); **H04L12/66** (20060101); **H04L67/12** (20220101)

U.S. Cl.:

CPC **H04L63/101** (20130101); **H04L12/66** (20130101); **H04L67/12** (20130101);

Background/Summary

TECHNICAL FIELD
[0001] The present invention relates to a communication control system, a gateway, and a communication control method.
BACKGROUND ART
[0002] As the risk of cyberattacks has increased in recent years, there has been a growing demand for precise access control using whitelists to protect vulnerable devices such as Internet of Things (IoT) devices from cyberattacks, including zero-day attacks, and efforts are being made to automatically generate whitelists for IoT devices and the like (see NPL 1).
CITATION LIST
Non Patent Literature
[0003] [NPL 1] Automatic generation and validation of controller whitelists (https://www.jstage.jst.go.jp/article/sicejl/60/1/60_21/_article/-char/ja/) [0004] [NPL 2] Permissions on Android (https://developer.android.com/guide/topics/permissions/overview)
Technical Problem
[0005] However, these efforts mainly involve generating whitelists afterwards from the normal state of already completed devices or apps, or blindly trusting and using whitelists defined by the manufacturer, and it is not possible for a third party to verify the accuracy of the whitelists.
[0006] Meanwhile, in recent years, attention has been focused on mechanisms that allow a third party to check whether purchased apps or devices have unnecessary permissions or engage in unnecessary communications unintended by a user (see NPL 2).
[0007] The present invention has been made in consideration of the above, and an object of the present invention is to provide a communication control system, a gateway, and a communication control method that allow a third party to check whether or not unnecessary communications unintended by a user are being performed.
Solution to Problem
[0008] In order to solve the above-mentioned problems and achieve the object, according to the present invention, there is provided a communication control system including a gateway and an IoT device connected to the gateway, wherein the gateway includes a request module configured to request, from the IoT device, information regarding normal communication performed by the IoT device, and a creation module configured to create a whitelist in which details of communication permitted for each IoT device are designated on a basis of the information regarding normal communication performed by the IoT device, which is transmitted from the IoT device, and the IoT device includes a transmission module configured to transmit the information regarding normal communication performed by the IoT device to the gateway in response to a request from the request module.
Advantageous Effects of Invention
[0009] According to the present invention, it is possible for a third party to verify whether or not communications that are inconvenient for a user are being performed.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0010] FIG. **1** is a block diagram illustrating an example of a configuration of a communication control system.

[0011] FIG. **2** is a block diagram illustrating an example of a configuration of a gateway.

[0012] FIG. **3** is a block diagram illustrating an example of a configuration of an IoT device.

[0013] FIG. **4** is a diagram for describing an overview of processing of the communication control system.

[0014] FIG. **5** is a diagram showing an example of information that defines normal communication performed by the IoT device.

[0015] FIG. **6** is a flowchart showing an example of a flow of processing performed by the communication control system.

DESCRIPTION OF EMBODIMENTS

[0016] Hereinafter, embodiments of a communication control system, a gateway, and a communication control method according to the present application will be described in detail with reference to the drawings. The present invention is not limited to the present embodiment. In addition, in the drawings, the same parts are denoted by the same reference numerals, and duplicate descriptions will be omitted.

[Overview of Present Invention]

[0017] In a communication control system **1** according to the present embodiment, a gateway **100** creates a whitelist on the basis of information regarding normal communication performed by an IoT device **200**, thereby enabling verification as to whether or not communications that are inconvenient for a user of the IoT device **200** are being performed.

[0018] [Configuration of Communication Control System] First, the configuration of the communication control device will be described with reference to FIG. **1**. As illustrated in FIG. **1**, the communication control system **1** includes a gateway **100** and an IoT device **200**. The gateway **100** accommodates the IoT device **200** and creates a whitelist on the basis of information regarding normal communication performed by the IoT device **200**, such as an NSTO.

[0019] Here, a Network Secure Transparency Object (NSTO) is information regarding normal communication performed by the IoT device **200**. The NSTO includes information such as, for example, a device ID, an IP address, a MAC address, a communication ID, a protocol, a port number, a destination, content included, encoding, a communication ID that should exist beforehand, a communication ID that should exist immediately beforehand, a communication ID that should exist afterward, and a communication ID that should exist immediately afterward.

[0020] The NSTO is defined by the manufacturer or the like of the IoT device **200** and checked by the user. For example, the NSTO is distributed in a form attached to the IoT device **200**, and the user verifies the NSTO by visual inspection or the like to check whether or not unnecessary communication is permitted. The NSTO format may be, for example, json, yaml, or xml.

[0021] The IoT device **200** is a device compatible with an NSTO or a terminal on which a predetermined application is installed. The IoT device **200** transmits information regarding normal communication performed by itself to the gateway **100**.

[0022] [Configuration of Gateway] Next, the configuration of the gateway **100** will be described with reference to FIG. **2**. As illustrated in FIG. **2**, the gateway **100** includes a communication module **110**, a control module **120**, and a storage module **130**. Note that each of these modules may be distributed and held in the plurality of devices. The processing of each of these modules will be described below.

[0023] The communication module **110** is realized by a Network Interface Card (NIC) or the like, and enables communication between an external device and the control module **120** via a telecommunication line such as a Local Area Network (LAN) or the Internet. For example, the communication module **110** enables the communication between the external device and the control module **120**.

[0024] The storage module **130** is realized by, for example, a semiconductor memory element such as a Random Access Memory (RAM) or a Flash Memory, or a storage device such as a hard disk or an optical disc. The information stored in the storage module **130** includes for example, an NSTO, a device ID, an IP address, a MAC address, a communication ID, a protocol, a port number, a destination, content included, encoding, a communication ID that should exist beforehand, a communication ID that should exist immediately beforehand, a communication ID that should exist afterward, a communication ID that should exist immediately afterward, and other information necessary for selection in communication control. Note that, the information stored in the storage module **130** is not limited to the above description.

[0025] The control module **120** is realized using a Central Processing Unit (CPU), a Network Processor (NP), a Field Programmable Gate Array (FPGA), or the like, and executes a processing program stored in a memory. As illustrated in FIG. **2**, the control module **120** includes a request module **121**, a creation module **122**, and a setting module **123**. Hereinafter, each module of the control module **120** will be described.

[0026] The request module **121** requests, from the IoT device **200**, information regarding normal communication performed by the IoT device **200**. For example, the request module **121** requests, from the IoT device **200**, an NSTO as information regarding normal communication performed by the IoT device **200**. To give a more specific example, the request module **121** requests information such as information such as a communication ID, a protocol, a port number, a destination, content included, encoding, a communication ID that should exist beforehand, a communication ID that should exist immediately beforehand, a communication ID that should exist afterward, and a communication ID that should exist immediately afterward.

[0027] Also, for example, upon receiving a request to start communication from a start request module **221** (described later), the request module **121** requests, from the IoT device **200**, information regarding normal communication performed by the IoT device **200**. For example, upon receiving a request to start communication from the start request module **221**, the request module **121** requests, from the IoT device **200**, transmission of an NSTO as information regarding normal communication performed by the IoT device **200**. Also, for example, the request module **121** receives a packet conveying a request to start communication from the start request module **221**, and transmits, to the IoT device **200**, a packet conveying a request for information regarding normal communication performed by the IoT device **200**.

[0028] The creation module **122** creates a whitelist in which details of communication permitted for each IoT device **200** are designated on the basis of information regarding normal communication performed by the IoT device **200** transmitted by the transmission module **222**. For example, the creation module **122** creates a whitelist in which details of communication permitted for each IoT device **200** are designated on the basis of the NSTO transmitted by the transmission module **222**.

[0029] To give a more specific example, the creation module **122** creates a whitelist for the IoT device **200** using information such as the communication ID, the protocol, the port number, the destination, the content included, the encoding, the communication ID that should exist beforehand, the communication ID that should exist immediately beforehand, the communication ID that should exist afterward, and the communication ID that should exist immediately afterward, which are transmitted from the IoT device **200**. Accordingly, the creation module **122** creates a whitelist in which only communications that are defined based on information regarding normal communication performed by the IoT device **200** are permitted.

[0030] The setting module **123** sets the whitelist created by the creation module **122**. For example, the setting module **123** sets the gateway **100** to permit only communications within the range described in the whitelist created by the creation module **122**. At this time, the setting module **123** may notify the IoT device **200** that the whitelist has been set, at the same time as setting the whitelist created by the creation module **122**.

[0031] [Configuration of IoT Device] Next, the configuration of the IoT device **200** will be described with reference to FIG. **3**. As illustrated in FIG. **3**, the IoT device **200** includes a communication module **210**, a control module **220**, and a storage module **230**. Note that each of these modules may be distributed and held in the plurality of devices. The processing of each of these modules will be described below.

[0032] The communication module **210** is realized by a Network Interface Card (NIC) or the like, and enables communication between an external device and the control module **120** via a telecommunication line such as a Local Area Network (LAN) or the Internet. For example, the communication module **210** enables the communication between the external device and the control module **220**.

[0033] The storage module **230** is realized by, for example, a semiconductor memory element such as a Random Access Memory (RAM) or a Flash Memory, or a storage device such as a hard disk or an optical disc. The information stored in the storage module **230** includes for example, an NSTO, a device ID, an IP address, a MAC address, a communication ID, a protocol, a port number, a destination, content included, encoding, a communication ID that should exist beforehand, a communication ID that should exist immediately beforehand, a communication ID that should exist afterward, a communication ID that should exist immediately afterward, and other information necessary for selection in communication control. Note that, the information stored in the storage module **230** is not limited to the above description.

[0034] The control module **220** is realized using a Central Processing Unit (CPU), a Network Processor (NP), a Field Programmable Gate Array (FPGA), or the like, and executes a processing program stored in a memory. As illustrated in FIG. **3**, the control module **220** includes a start request module **221** and a transmission module **222**. Hereinafter, each module of the control module **220** will be described.

[0035] The start request module **221** requests the gateway **100** to start communication. For example, the start request module **221** transmits a packet conveying a request to start communication to the gateway **100**.

[0036] In response to a request from the request module **121**, the transmission module **222** transmits information regarding normal communication performed by the IoT device **200** to the gateway **100**. For example, in response to a request from the request module **121**, the transmission module **222** transmits an NSTO as information regarding normal communication performed by the IoT device **200** to the gateway **100**. To give a more specific example, in response to a request from the request module **121**, the transmission module **222** transmits, to the gateway **100**, information such as the communication ID, the protocol, the port number, the destination, the content included, the encoding, the communication ID that should exist beforehand, the communication ID that should exist immediately beforehand, the communication ID that should exist afterward, and the communication ID that should exist immediately afterward.

[0037] [Overview of Processing] Next, an overview of the processing performed by the communication control system **1** will be described with reference to FIG. **4**. First, the start request module **221** of the IoT device **200** sends a request to start communication to the gateway **100**. Next, upon receiving a request to start communication from the start request module **221**, the gateway **100** requests, from the IoT device **200**, information regarding normal communication performed by the IoT device **200**. At this time, the gateway **100** may notify the IoT device **200** of version information and the like at the same time as requesting information regarding normal communication performed by the IoT device **200**.

[0038] Then, upon receiving the request from the request module **121**, the transmission module **222** of the IoT device **200** transmits information regarding normal communication performed by the IoT device **200** to the gateway **100**. Thereafter, the creation module **122** of the gateway **100** creates a whitelist using information regarding normal communication performed by the IoT device **200**. For example, the creation module **122** creates a whitelist in which details of communication permitted for each IoT device **200** are designated on the basis of the NSTO as information regarding normal communication performed by the IoT device **200** transmitted by the transmission module **222**. Then, the setting module **123** sets the whitelist created by the creation module **122**.

[0039] Accordingly, the communication control system **1** not only ensures that only communications intended by the manufacturer are performed through access control by the gateway **100** using information regarding normal communication performed by the IoT device **200**, but also makes it possible to verify whether or not communications that are inconvenient for the user are being performed.

[0040] Next, information regarding normal communication performed by the IoT device **200** will be described with reference to FIG. **5**. The information regarding normal communication performed by the IoT device **200** requested by the request module **121** of the gateway **100** includes information such as the communication ID, the protocol, the port number, the destination, the content, the encoding, the communication ID that should exist beforehand, the communication ID that should exist immediately beforehand, the communication ID that should exist afterward, and the communication ID that should exist immediately afterward. The details described in the content may differ depending on the type of protocol. For example, the details described in the content may be a set of regular expressions that a value in a certain field has to satisfy.

[0041] Thereby, the communication control system **1** makes it easy for a third party to examine the details of the communication performed by the IoT device **200**.

[0042] [Flowchart] Next, the flow of processing performed by the gateway **100** will be described with reference to FIG. **6**. Note that the following steps S**101** to S**106** may be executed in a different order. Furthermore, among the following steps S**101** to S**106**, some processes may be omitted.

[0043] First, the start request module **221** of the IoT device **200** makes a request to start communication to the gateway **100** (step S**101**). For example, the start request module **221** of the IoT device **200** transmits a packet conveying a request to start communication to the gateway **100**.

[0044] Next, upon receiving the request from the start request module **221**, the request module **121** of the gateway **100** requests, from the IoT device **200**, information regarding normal communication performed by the IoT device **200** (step S**102**). For example, the request module **121** of the gateway **100** requests, from the IoT device **200**, transmission of an NSTO as information regarding normal communication performed by the IoT device **200**.

[0045] Then, upon receiving the request from the request module **121**, the transmission module **222** of the IoT device **200** transmits information regarding normal communication performed by the IoT device **200** to the gateway **100** (step S**103**). For example, upon receiving the request from the request module **121**, the transmission module **222** of the IoT device **200** transmits an NSTO as information regarding normal communication performed by the IoT device **200** to the gateway **100**.

[0046] Subsequently, the creation module **122** of the gateway **100** creates a whitelist in which details of communication permitted for each IoT device **200** are designated on the basis of information regarding the communication transmitted by the transmission module **222** (step S**104**). For example, the creation module **122** of the gateway **100** creates a whitelist in which details of communication permitted for each IoT device **200** are designated on the basis of the NSTO as information regarding normal communication performed by the IoT device **200** transmitted by the transmission module **222**.

[0047] Thereafter, the setting module **123** of the gateway **100** sets the whitelist created by the creation module **122** (step S**105**). For example, the setting module **123** sets the gateway **100** to permit only communications within the range described in the whitelist created by the creation module **122**. At this time, for example, the setting module **123** of the gateway **100** may notify the IoT device **200** that the created whitelist has been set.

[0048] Then, with the whitelist set by the setting module **123**, the IoT device **200** is permitted to communicate within the range described in the whitelist (step S**106**). For example, communication within a range described in a whitelist set by the setting module **123** of the IoT device **200** is permitted, and communication outside the range described in the whitelist is not permitted.

[0049] [Effects] The communication control system **1** according to the embodiment is a communication control system including a gateway **100** and an IoT device **200** connected to the gateway **100**, in which the gateway **100** includes a request module **121** that requests, from the IoT device **200**, information regarding normal communication performed by the IoT device **200**, and a creation module **122** that creates a whitelist in which details of communication permitted for each IoT device **200** are designated on the basis of the information regarding normal communication performed by the IoT device **200**, which is transmitted from the IoT device **200**, and the IoT device **200** includes a transmission module **222** that transmits the information regarding normal communication performed by the IoT device **200** to the gateway **100** in response to a request from the request module **121**.

[0050] Accordingly, the communication control system **1** enables the gateway **100** to use information regarding normal communication performed by the IoT device **200** to create a whitelist, thereby enabling a third party to verify whether or not access permissions that are inconvenient for the user are being granted.

[0051] In the communication control system **1** according to the embodiment, the request module **121** of the gateway **100** requests a Network Secure Transparency Object (N STO) as the information regarding normal communication performed by the IoT device **200**.

[0052] Accordingly, the communication control system **1** enables the gateway **100** to use the NSTO as information regarding normal communication performed by the IoT device **200** to create a whitelist, thereby enabling a third party to verify whether or not access permissions that are inconvenient for the user are being granted.

[0053] In the communication control system **1** according to the embodiment, the gateway **100** further includes a setting module **123** that sets the whitelist created by the creation module **122**.

[0054] Accordingly, the communication control system **1** enables the gateway **100** to use information regarding normal communication performed by the IoT device **200** to create a whitelist, thereby enabling a third party to verify whether or not access permissions that are inconvenient for the user are being granted.

[0055] In the communication control system **1** according to the embodiment, the IoT device **200** further includes a start request module **221** that requests the gateway **100** to start communication, and the request module **121** of the gateway **100** receives a request to start communication from the start request module **221** and requests, from the IoT device **200**, the information regarding normal communication performed by the IoT device **200**.

[0056] Accordingly, the communication control system **1** enables the gateway **100** to use information regarding normal communication performed by the IoT device **200** to create a whitelist, thereby enabling a third party to verify whether or not access permissions that are inconvenient for the user are being granted.

[0057] The gateway **100** according to the embodiment includes a request module **121** that requests, from the IoT device **200**, information regarding normal communication performed by the IoT device **200**, and a creation module **122** that creates a whitelist in which details of communication permitted for each IoT device **200** are designated on the basis of the information regarding normal communication performed by the IoT device **200**, which is transmitted from the IoT device **200**.

[0058] This enables the gateway **100** to use information regarding normal communication performed by the IoT device **200** to create a whitelist, thereby enabling a third party to verify whether or not access permissions that are inconvenient for the user are being granted.

[0059] [System Configuration, Etc.] In addition, each component of each device that has been illustrated is functionally conceptual, and is not necessarily physically configured as illustrated. In other words, the specific aspects of distribution and integration of the devices are not limited to those illustrated in the drawings, all or part of the components may be distributed or integrated functionally or physically in desired units depending on various kinds of loads and states of use. For example, all or any part of the processing functions performed by the devices may be achieved by a CPU and programs analyzed and executed by the CPU or achieved as hardware by a wired logic.

[0060] Further, among the processes described in the present embodiment, all or some of the processes described as being automatically performed can also be manually performed, or all or some of the processes described as being manually performed can also be performed automatically using a known method. In addition, the processing procedure, the control procedure, specific names, information including various types of data and parameters that are shown in the above document and drawings may be arbitrarily changed unless otherwise described.

[0061] [Others] While various embodiments have been described in detail in the present specification with reference to the drawings, a plurality of these embodiments are examples and the present invention is not intended to limit to the plurality of these embodiments. The features described in the present specification can be realized according to various methods, including various modifications and improvements based on the knowledge of those skilled in the art.

[0062] In addition, the above-mentioned "module (-er suffix, -or suffix)" can be read as a unit, means, a circuit, or the like. For example, the communication module, the control module, and the storage module can be read as a communication unit, a control unit, and a storage unit, respectively.

REFERENCE SIGNS LIST

[0063] **1** Communication control system [0064] **100** Gateway [0065] **110** Communication module [0066] **120** Control module [0067] **121** Request module [0068] **122** Creation module [0069] **123** Setting module [0070] **130** Storage module [0071] **200** IoT device [0072] **210** Communication module [0073] **220** Control module [0074] **221** Start request module [0075] **222** Transmission module [0076] **230** Storage module

Claims

1. A communication control system comprising a gateway and an IoT device connected to the gateway, wherein the gateway includes: a request module configured to request, from the IoT device, information regarding normal communication performed by the IoT device; and a creation module configured to create a whitelist in which details of communication permitted for each IoT device are designated on a basis of the information regarding normal communication performed by the IoT device, which is transmitted from the IoT device, and the IoT device includes a transmission module configured to transmit the information regarding normal communication performed by the IoT device to the gateway in response to a request from the request module.

2. The communication control system according to claim 1, wherein the request module requests a Network Secure Transparency Object (NSTO) as the information regarding normal communication performed by the IoT device.

3. The communication control system according to claim 1, wherein the gateway further includes a setting module configured to set the whitelist created by the creation module.

4. The communication control system according to claim 1, wherein the IoT device further includes a start request module configured to request the gateway to start communication, and the request module receives a request to start communication from the start request module and requests, from the IoT device, the information regarding normal communication performed by the IoT device.

5. A gateway comprising: a request module configured to request, from an IoT device, information regarding normal communication performed by the IoT device; and a creation module configured to create a whitelist in which details of communication permitted for each IoT device are designated on a basis of the information regarding normal communication performed by the IoT device, which is transmitted from the IoT device.

6. A communication control method executed by a gateway and an IoT device connected to the gateway, the communication control method comprising: a request step in which the gateway requests, from the IoT device, information regarding normal communication performed by the IoT device; a transmission step in which the IoT device transmits the information regarding normal communication performed by the IoT device to the gateway in response to a request from the request step; and a creation step in which the gateway creates a whitelist in which details of communication permitted for each IoT device are designated on a basis of the information regarding normal communication performed by the IoT device, which is transmitted in the transmission step.